

Solving Word Problems Involving Volume of Pyramids

Jonathan R. Bacolod

The **volume** of any pyramid is one-third the volume of a prism with the same base and height.

Recall: Volume Formula

$$V = \frac{1}{3}Bh$$

B is the **area of the base** and *h* is the **perpendicular height**.

Recall: Base Area

For a square base:

$$B = s^2$$

For a rectangular base:

$$B = lw$$

Key Words

These questions ask for the **volume** of a pyramid:

- ▶ “What is its volume?”
- ▶ “How much soil can it hold?”
- ▶ “How many cubic centimeters does it contain?”

Steps in Solving Word Problems

Read the problem carefully.

Identify the **given**
measurements.

Compute the **base** area first.

Apply the **volume** formula.

Write the answer in cubic **units**.

Memory Tip

R-G-B

Read **G**iven **B**ase area

Then find the **V**olume and write the **U**nits.

Cubic Units

Always write the final answer in **cubic units**.

Length	Volume
cm	cm^3
m	m^3
in	in^3

Examples

Example 1

A square pyramid has a base with side length 9 cm and a height of 10 cm. Find its volume.

$$B = s^2 = (9)^2 = 81 \text{ cm}^2$$

$$\begin{aligned} V &= \frac{1}{3}Bh \\ &= \frac{1}{3}(81)(10) \\ &= 270 \text{ cm}^3 \end{aligned}$$

Example 2

A rectangular pyramid has a base measuring 12 cm by 5 cm and a height of 9 cm. Find its volume.

$$B = lw = (12)(5) = 60 \text{ cm}^2$$

$$\begin{aligned} V &= \frac{1}{3}Bh \\ &= \frac{1}{3}(60)(9) \\ &= 180 \text{ cm}^3 \end{aligned}$$

Example 3

A sandbox is shaped like a rectangular pyramid with a base measuring 6 m by 4 m and a height of 3 m. How much sand can it hold?

$$B = lw = (6)(4) = 24 \text{ m}^2$$

$$\begin{aligned} V &= \frac{1}{3}Bh \\ &= \frac{1}{3}(24)(3) \\ &= 24 \text{ m}^3 \end{aligned}$$

Example 4

A tent is shaped like a square pyramid. The base side is 4 m, and the height is 3 m. What is its volume?

$$B = s^2 = (4)^2 = 16 \text{ m}^2$$

$$\begin{aligned} V &= \frac{1}{3}Bh \\ &= \frac{1}{3}(16)(3) \\ &= 16 \text{ m}^3 \end{aligned}$$

Practice Exercises

Practice 1

A square pyramid has a base with side length 12 cm and a height of 15 cm. Find its volume.

Practice 2

A rectangular pyramid has a base measuring 10 cm by 8 cm and a height of 18 cm. Find its volume.

Practice 3

A square pyramid has a base side of 14 m and a height of 9 m. Determine its volume.

Practice 4

A rectangular pyramid has a base measuring 16 m by 9 m and a height of 12 m. Find its volume.

Practice 5

A monument is shaped like a square pyramid. The base side is 25 m, and the height is 18 m. Find the volume.

Practice 6

A flower bed is shaped like a rectangular pyramid with a base measuring 7 m by 4 m and a height of 6 m. How much soil can it hold?

Practice 7

A solid toy is shaped like a square pyramid with a base side of 9 cm and a height of 8 cm. Find its volume.

Activities

Activity 1

A square pyramid has a base with side length 15 cm and a height of 12 cm. Find its volume.

Activity 2

A rectangular pyramid has a base measuring 12 cm by 9 cm and a height of 15 cm. Find its volume.

Activity 3

A square pyramid has a base side of 16 m and a height of 15 m. Determine its volume.

Activity 4

A rectangular pyramid has a base measuring 18 m by 10 m and a height of 14 m. Find its volume.

Activity 5

A pyramid has a rectangular base measuring 24 cm by 18 cm and a height of 9 cm. How many cubic centimeters does it contain?

Activity 6

A decorative pyramid has a square base with side length 20 m and a height of 24 m. Find its volume.

Activity 7

A storage container is shaped like a rectangular pyramid. Its base measures 8 m by 6 m, and its height is 10 m. What is its volume?

Thank you!